

Awesome! Here is the full presentation script for the next speaker:

Speaker 8: Susanta

Topic: Tech Stack – What tools and technologies did we use and why?

Script (Presentation Story)

Thank you, Puja.

Now I'll explain the **tech stack** we used to build MessFinder – in simple terms, this means the set of tools, technologies, and platforms we used for both frontend and backend.

We selected these tools based on four important factors:

- Simplicity
- Scalability
- Performance
- Cost-effectiveness

Let's break down each component:

Frontend – React.js

We used **React** for building the user interface. It's a modern JavaScript library developed by Facebook.

Why React?

- It's **component-based**, which means we can reuse code easily
- It allows us to create **dynamic and fast interfaces**
- It works great with state management and routing

We paired React with **Vite** instead of CRA, for faster development.

Styling – Tailwind CSS

For design, we used **Tailwind CSS**, which is a utility-first CSS framework.

Why Tailwind?

- It allows **mobile-first, responsive design**
- Makes styling **faster and easier** with ready-made classes
- Helps us avoid writing bulky custom CSS

This helped us make the app look clean and load faster across all devices.

Routing – React Router

We used **React Router** to manage navigation between pages like Home, Room Details, Submit, Login, etc.

It helps us create **dynamic URLs**, such as:

```
/details/:roomId
```

This gives us the flexibility to open unique room listings with clean links.

Backend – Firebase

Instead of building our own backend server, we used **Firebase**, which is a Backend-as-a-Service offered by Google.

It gives us:

- **Firestore** – a real-time NoSQL database
- **Firebase Auth** – to manage login securely
- **Cloud Storage** – to store images uploaded by owners

Firebase made it possible to **build a full-stack app without writing backend code**.

Authentication – Firebase Auth

For login and registration, we used Firebase Auth.

It supports:

- Email/password login
- Google login
- Automatic session management

This made our app secure and easy to use for both users and owners.

Database – Firestore

We used **Firestore**, which is Firebase’s NoSQL cloud database.

It stores data in **collections and documents**, which is very flexible and fast for real-time apps.

We used it to store:

- Room listings
 - User profiles
 - Chat messages
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Cloud Storage – Firebase Storage

To upload and store images of rooms, we used Firebase Storage.

- Owners can upload photos
 - URLs are stored in Firestore
 - Access is secured via rules and tokens
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Hosting – Vercel + Firebase

We hosted our frontend on **Vercel**, which supports React out of the box. It’s free, fast, and integrates with GitHub for auto deployment.

Firebase handles backend hosting, database, and media — making our architecture **completely serverless**.

Summary:

- **React + Tailwind** = fast and beautiful UI
- **Firebase** = secure backend, auth, DB, and storage
- **Vercel** = simple and scalable hosting

We didn’t just choose random tools — we selected a stack that was **modern, easy to work with, and powerful enough to build a real solution**.

Now, I’d like to invite **Aakash**, who will explain the **User Roles and Application Flow** in more detail.

Tips for Susanta:

- Speak confidently about tools – as if you tested and chose them yourself
 - Pause briefly after each tool name
 - Don't list everything like a dictionary – connect each tool to your app's needs
 - End clearly: "Now I'd like to invite..."
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Likely Viva/Panel Questions for Susanta:

Q: Why did you not use MongoDB or Node.js?

A: Firebase already gave us authentication, database, and hosting – so we didn't need to set up a separate backend like Node or Mongo.

Q: What's the advantage of using Vercel over Firebase Hosting?

A: Vercel offers faster build times for React and integrates smoothly with GitHub for CI/CD.

Q: Are there any limits with Firebase?

A: Yes – Firestore has limits on queries and reads per day. But for student-level projects or startups, it's sufficient.
